RECALLING Article 28(b) of the Convention on the International Maritime Organization con
cerning the functions of the Committee, RECALLING ALSO resolution MSC.268(85) by which i
t adopted the International Maritime Solid Bulk Cargoes Code (hereinafter referred to as
"the IMSBC Code"), which has become mandatory under chapter VI of the International Conv
ention for the Safety of Life at Sea, 1974, as amended ("the Convention"), RECALLING FUR
THER article VIII(b) and regulation VII/1.1 of the Convention concerning the procedure f
or amending the IMSBC Code, HAVING CONSIDERED, at its 105th session, amendments to the I
MSBC Code, proposed and --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentr
e/IndexofIMOResolutions/MSCResolutions/MSC.500(105).pdf procedure for amending the IMSBC
Code, HAVING CONSIDERED, at its 105th session, amendments to the IMSBC Code, proposed an
d circulated in accordance with article VIII(b)(i) of the Convention, 1 ADOPTS, in accor
dance with article VIII(b)(iv) of the Convention, amendments to the IMSBC Code, the text
of which is set out in the annex to the present resolution; 2 DETERMINES, in accordance
with article VIII(b)(vi)(2)(bb) of the Convention, that the said amendments shall be dee
med to have been accepted on 1 June 2023, unless prior to that date more than one third
of the Contracting Governments to the Convention or Contracting Governments the combined
merchant fleets of which constitute not less than 50% of the gross tonnage of the world'
s merchant fleet have notified their objections to the amendments; 3 INVITES Contracting
Governments to the Convention to note that, in accordance with article VIII(b)(vii)(2)
of the Convention, the amendments shall enter into force --- URL: https://wwwcdn.imo.org
/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.500(105).pdf
article VIII(b)(vii)(2) of the Convention, the amendments shall enter into force on 1 D
ecember 2023 upon their acceptance in accordance with paragraph 2 above; 4 AGREES that C
ontracting Governments to the Convention may apply the aforementioned amendments in whol
e or in part on a voluntary basis as from 1 January 2023; 5 REQUESTS the Secretary -Gene
ral, for the purpose of article VIII(b)(v) of the Convention, to transmit certified copi
es of the present resolution and the text of the amendments contained in the annex to al
l Contracting Governments to the Convention; 6 ALSO REQUESTS the Secretary-General to tr
ansmit copies of this resolution and its annex to Members of the Organization which are
not Contracting Governments to the Convention.

RESOLUTION MSC.500(105) (adopted on 28 April 2022) AMENDMENTS TO THE INTERNATIONAL MARIT
IME SOLID BULK CARGOES (IMSBC) CODE - 2 - ANNEX AMENDMENTS TO THE INTERNATIONAL MARITIME
SOLID BULK CARGOES (IMSBC) CODE --- URL: https://wwwcdn.imo.org/localresources/en/Knowle
dgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.500(105).pdf - 2 - ANNEX AMENDMENTS T
O THE INTERNATIONAL MARITIME SOLID BULK CARGOES (IMSBC) CODE Table of contents For "Sect
ion 7", replace the text to read "Cargoes which may liquefy or undergo dynamic separatio
n".

For "Section 8", replace the text to read "Test procedures for group A cargoes".

Section 1 General provisions 1.7 Definitions In the definition for " GHS", replace the w
ord "seventh" with the word "ninth" and replace "ST/SG/AC.10/30/Rev.7" with "ST/SG/AC.10
/30/Rev.9".

In the definition for " Group A", replace the text to read " Group A consists of cargoes
which possess a hazard due to moisture that may result in liquefaction or dynamic separa
tion if shipped at a moisture content in excess of their transportable moisture limit."
In the definition for "Group C", replace the text to read "Group C consists of cargoes w
hich are classified as neither group A nor group B." --- URL: https://wwwcdn.imo.org/loc
alresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.500(105).pdf In
the definition for "Group C", replace the text to read "Group C consists of cargoes whic
h are classified as neither group A nor group B." In the definition for " Manual of Test
s and Criteria ", replace the definition to read " Manual of Tests and Criteria means th
e seventh revised edition of the United Nations publication entitled Manual of Tests and
Criteria (ST/SG/AC.10/11/Rev.7 and Amend.1)." In the definition for "Transportable moist
ure limit (TML)", replace the text to read "Transportable moisture limit (TML) of a grou
p A cargo means the maximum moisture content of the cargo which is considered safe for c
arriage in ships not complying with the special provisions of 7.3.2.

It is determined by the test procedures, approved by a competent authority, such as thos
e specified in paragraph 1 of appendix 2." Add the following new definitions in alphabet
ical order: "Cargoes which may undergo dynamic separation means cargoes which contain a
certain --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOReso
lutions/MSCResolutions/MSC.500(105).pdf Add the following new definitions in alphabetica
l order: "Cargoes which may undergo dynamic separation means cargoes which contain a cer

tain proportion of fine particles and a certain amount of moisture, and may undergo dyna
mic separation if shipped at a moisture content in excess of their transportable moistur
e limit." "Dynamic separation means the phenomenon of forming a liquid slurry (water and
fine solids) above the solid material, resulting in a free surface effect which may sign
ificantly affect the ship's stability." RESOLUTION MSC.500(105) (adopted on 28 April 202
2) AMENDMENTS TO THE INTERNATIONAL MARITIME SOLID BULK CARGOES (IMSBC) CODE - 3 - Sectio
n 4 Assessment of acceptability of consignments for safe shipment 4.1 Identification and
classification In 4.1.4, replace the words "Manual of Tests and Criteria" with the words
"Manual of Tests and Criteria".

4.2 Provision of information In 4.2.2.9, replace the existing text to read as follows: -
-- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/M
SCResolutions/MSC.500(105).pdf Criteria".

4.2 Provision of information In 4.2.2.9, replace the existing text to read as follows: "
additional information in the form of a certificate on the moisture content of the cargo
and its transportable moisture limit in the case of a group A cargo;".

In 4.2.3, in the table for "Form for cargo information for solid bulk cargoes", in the "
Group of the cargo" box, delete the asterisks and the corresponding footnote text "For c
argoes which may liquefy (group A and group A and B cargoes)".

Under "Transportable moisture limit", add the words "For group A and B and group A cargo
es".

Under "Moisture content at shipment", add the words "For group A and B and group A cargo
es".

4.3 Certificates of test In 4.3.2, replace the first sentence to read as follows: "When
a group A cargo is carried, the shipper shall provide the ship's master or his represent
ative with a signed certificate of the TML, and a signed certificate or --- URL: https:/
/wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/M
SC.500(105).pdf representative with a signed certificate of the TML, and a signed certif
icate or declaration of the moisture content, each issued by an entity recognized by the
competent authority of the port of loading." In 4.3.3, replace the first sentence to rea
d as follows: "When a group A cargo is carried, procedures for sampling, testing and con
trolling moisture content to ensure the moisture content is less than the TML when it is
on board the ship shall be established by the shipper, taking account of the provisions
of this Code." In 4.3.5, replace the first sentence to read as follows: "When a group A
cargo is to be loaded into more than one cargo space of a ship, the certificate or the d
eclaration of moisture content shall certify the moisture content of each type of finely
grained material loaded into each cargo space." 4.4 Sampling procedures In 4.4.3, replac
e the existing text with the following: --- URL: https://wwwcdn.imo.org/localresources/e
n/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.500(105).pdf 4.4 Sampling pro
cedures In 4.4.3, replace the existing text with the following: "For a group A cargo, th
e shipper shall facilitate access to stockpiles for the purpose of inspection, sampling
and subsequent testing by the ship's nominated representative." 4.8 Documentation requir
ed on board the ship carrying dangerous goods In 4.8.3, after the words "SOLAS regulatio
n II-2/19.4", add the words "(or II-2/54.3)".

RESOLUTION MSC.500(105) (adopted on 28 April 2022) AMENDMENTS TO THE INTERNATIONAL MARIT
IME SOLID BULK CARGOES (IMSBC) CODE - 4 - Section 7 Cargoes which may liquefy In the hea
ding for "Section 7", replace the words "Cargoes which may liquefy" with the words "Carg
oes which may liquefy or undergo dynamic separation".

7.1 Introduction In 7.1.1, replace the existing text with the following: "The purpose of
this section is to bring to the attention of masters and others with --- URL: https://ww
wcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.
500(105).pdf In 7.1.1, replace the existing text with the following: "The purpose of thi
s section is to bring to the attention of masters and others with responsibilities for t
he loading and carriage of bulk cargoes, the risks associated with liquefaction or dynam
ic separation and the precautions to minimize the risk.

Such cargoes may appear to be in a relatively dry granular state when loaded, and yet ma
y contain sufficient moisture to become fluid or unstable under the stimulus of compacti
on and the vibration which occurs during a voyage." In 7.1.3, replace the existing text
with the following: "Some group A cargoes may also heat spontaneously." 7.3 Provisions f
or cargoes that may liquefy In the heading for 7.3, replace the words "Provisions for ca
rgoes that may liquefy" with the words "Provisions for cargoes which may liquefy or unde
rgo dynamic separation".

7.3.1 General In 7.3.1.1, replace the existing text with the following: --- URL: https:/
/wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/M
SC.500(105).pdf 7.3.1 General In 7.3.1.1, replace the existing text with the following:
"Group A cargoes shall only be accepted for loading when the actual moisture content of
the cargo is less than its TML.

Notwithstanding this provision, cargoes having moisture content in excess of the TML may
be carried on a specially constructed or fitted cargo ship for confining cargo shift spe
cified in 7.3.2." In 7.3.1.2, replace the existing text with the following: "Notwithstan
ding the provisions in 1.4 of this Code, the requirements in 4.2.2.9, 4.2.2.10, 4.3.2 to
4.3.5, 4.5, 4.6 and 8 of this Code need not apply to a group A cargo provided that the c
argo is carried on a specially constructed or fitted cargo ship for confining cargo shif
t specified in 7.3.2 or on a specially constructed ship for dry powdery cargoes specifie
d in 7.3.3." Section 8 Test procedures for cargoes which may liquefy In the heading for
"Section 8", replace the words "Test procedures for cargoes which may --- URL: https://w
wwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC
.500(105).pdf Section 8 Test procedures for cargoes which may liquefy In the heading for
"Section 8", replace the words "Test procedures for cargoes which may liquefy" with the
words "Test procedures for group A cargoes".

RESOLUTION MSC.500(105) (adopted on 28 April 2022) AMENDMENTS TO THE INTERNATIONAL MARIT
IME SOLID BULK CARGOES (IMSBC) CODE - 5 - Section 9 Materials possessing chemical hazard
s 9.2 Hazard classification 9.2.3 Materials hazardous only in bulk (MHB) 9.2.3.2 Combust
ible solids: MHB (CB) In 9.2.3.2.2, replace the word "33.2.1.4.3.1" with the word "33.2.
4.3.1".

9.2.3.3 Self-heating solids: MHB (SH) In 9.2.3.3.2, replace the word "33.3.1.6" with the
word "33.4.6".

In 9.2.3.3.3, replace the word "33.4.1.4.3.5" with the word "33.5.4.3.5".

9.2.3.4 Solids that evolve flammable gas when wet: MHB (WF) In 9.2.3.4.2, replace the wo
rd "33.4.1" with the word "33.5".

9.2.3.5 Solids that evolve toxic gas when wet: MHB (WT) --- URL: https://wwwcdn.imo.org/
localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.500(105).pdf
In 9.2.3.4.2, replace the word "33.4.1" with the word "33.5".

9.2.3.5 Solids that evolve toxic gas when wet: MHB (WT) In 9.2.3.5.2, replace the word "
33.4.1" with the word "33.5".

9.2.3.7 Corrosive solids: MHB (CR) In 9.2.3.7.3, replace the existing text with the foll
owing: "A material shall be classified as MHB when the corros ion rate on a steel surfac
e is between 4 mm and 6.25 mm a year at a test temperature of 55°C.

For the purposes of testing steel, type S235JR+CR (1.0037 resp.

St 37 -2), S275J2G3+CR (1.0144 resp.

St 44 -3), ISO 3574:1999, Unified Numbering Systems (UNS) G1020 0 or SAE 1020 shall be u
sed.

An acceptable test is prescribed in the United Nations Manual of Tests and Criteria, par
t III, section 37.

A representative sample of the cargo shall be tested at typical as -shipped cargo condit
ions, including moisture content by mass, bulk density, particle size distribution and a
tmospheric conditions.

--- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/
MSCResolutions/MSC.500(105).pdf mass, bulk density, particle size distribution and atmos
pheric conditions.

The assessment of localized corrosion occurring upon the steel surface shall be performe
d using national or international standards.

The test shall be conducted taking into account the guidance developed by the Organizati
on.", APPENDIX 1 INDIVIDUAL SCHEDULES OF SOLID BULK CARGOES Amendments to existing indiv
idual schedules Amend the following individual schedules as indicated below: ALFALFA In
the individual schedule for "ALFALFA", in the section for "Characteristics", in the tabl
e, replace the words "Fine Powder" with the words "Fine powder".

RESOLUTION MSC.500(105) (adopted on 28 April 2022) AMENDMENTS TO THE INTERNATIONAL MARIT
IME SOLID BULK CARGOES (IMSBC) CODE - 6 - ALUMINIUM NITRATE UN 1438 In the individual sc
hedule for "ALUMINIUM NITRATE UN 1438", in the section for "Emergency procedures", in th
e table, replace the word "overalls" with the word "coveralls".

--- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.500(105).pdf procedures", in the table, replace the word "overalls" with the word "coveralls".

AMMONIUM NITRATE UN 1942 In the individual schedule for "AMMONIUM NITRATE UN 1942", in the section for "Description", in the "Note", replace the word " UN" by the word "UN".

In the section for "Loading", delete the word "intrinsically".

AMMONIUM NITRATE BASED FERTILIZER UN 2067 In the individual schedule for "AMMONIUM NITRATE BASED FERTILIZER UN 2067", in the section for "Loading", delete the word "intrinsically".

AMMONIUM NITRATE BASED FERTILIZER UN 2071 In the individual schedule for "AMMONIUM NITRATE BASED FERTILIZER UN 2071", under the BCSN, replace the word " UN" by the word "UN".

In the section for "Loading", delete the word "intrinsically".

AMMONIUM NITRATE BASED FERTILIZER (non-hazardous) Delete the individual schedule for "AMMONIUM NITRATE BASED FERTILIZER (non-hazardous)".

BARIUM NITRATE UN 1446 --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.500(105).pdf Delete the individual schedule for "AMMONIUM NITRATE BASED FERTILIZER (non-hazardous)".

BARIUM NITRATE UN 1446 In the individual schedule for "BARIUM NITRATE UN 1446", in the section for "Emergency procedures", in the table, replace the word "overalls" with the word "coveralls".

CASTOR BEANS or CASTOR MEAL or CASTOR POMACE or CASTOR FLAKE UN 2969 In the individual schedule for "CASTOR BEANS or CASTOR MEAL or CASTOR POMACE or CASTOR FLAKE UN 2969", move the sentence "Castor meal, castor pomace and castor flakes shall not be carried in bulk.
" from the section for "Precautions" to underneath the BCSN.

COAL TAR PITCH In the individual schedule for "COAL TAR PITCH", under the section for "Description", replace the words "Coal Tar" with the words "coal tar" and replace the word "Cokes" with the word "cokes".

CRUSHED CARBON ANODES In the individual schedule for "CRUSHED CARBON ANODES", under the section for --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.500(105).pdf "cokes".

CRUSHED CARBON ANODES In the individual schedule for "CRUSHED CARBON ANODES", under the section for "Description", replace the words "Crushed Car bon Anodes" with the words "Crushed carbon anodes".

FISH MEAL (FISH SCRAP), STABILIZED UN 2216 Anti-oxidant treated In the individual schedule for "FISH MEAL (FISH SCRAP), STABILIZED UN 2216 Anti-oxidant treated", under the BCSN, replace the word "fishmeal" with the words "fish meal".

RESOLUTION MSC.500(105) (adopted on 28 April 2022) AMENDMENTS TO THE INTERNATIONAL MARITIME SOLID BULK CARGOES (IMSBC) CODE - 7 - IRON ORE FINES In the individual schedule for "IRON ORE FINES", in the table for "Characteristics", regarding bulk density, replace the words "1,500 to 3,000" with the words "1,500 to 3,500".

METAL SULPHIDE CONCENTRATES, CORROSIVE UN 1759 In the individual schedule for "METAL SULPHIDE CONCENTRATES, CORROSIVE UN 1759", --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.500(105).pdf METAL SULPHIDE CONCENTRATES, CORROSIVE UN 1759 In the individual schedule for "METAL SULPHIDE CONCENTRATES, CORROSIVE UN 1759", under the BCSN, replace the words "Packing Group" with the words "packing group".

METAL SULPHIDE CONCENTRATES, SELF-HEATING UN 3190 In the individual schedule for "METAL SULPHIDE CONCENTRATES, SELF -HEATING UN 3190", under the BCSN, replace the words "Packing Group" with the words "packing group".

Mineral concentrates In the individual schedule for "Mineral concentrates", replace the words "NEFELENE SYENITE (mineral)" with the words "NEPHELINE SYENITE (mineral)".

PITCH PRILL In the individual schedule for " PITCH PRILL", in the section for "Emergency procedures", in the table, replace the word "overalls" with the word "coveralls".

SUPERPHOSPHATE (triple, granular) Delete the individual schedule for "SUPERPHOSPHATE (triple, granular)".

SYNTHETIC CALCIUM FLUORIDE --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.500(105).pdf SUPERPHOSPHATE (triple, granular) Delete the individual schedule for "SUPERPHOSPHATE (triple, granular)".

SYNTHETIC CALCIUM FLUORIDE In the individual schedule for "SYNTHETIC CALCIUM FLUORIDE", under the section for "Description", replace "70%-80%" with the words "70% to 80%", repl ace "5% -10%" with the words "5% to 10%" and replace "10%-20%" with the words "10% to 20 %".

New individual schedules Insert the following new individual schedules in alphabetical o rder: "AMMONIUM NITRATE BASED FERTILIZER This schedule shall only apply to ammonium nitr ate based fertilizers which do not meet any of the criteria on dangerous goods or materi als hazardous only in bulk specified in 9.2.2 or 9.2.3 of this Code, respectively.

Ammonium nitrate based fertilizers transported in conditions mentioned in this schedule are straight nitrogen fertilizers or compound fertilizers within the following compositi on limits: --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOR esolutions/MSCResolutions/MSC.500(105).pdf straight nitrogen fertilizers or compound fer tilizers within the following composition limits: Straight nitrogen fertilizers containi ng less than 2% chloride, and .1 not more than 70% ammonium nitrate with other inorganic materials; or RESOLUTION MSC.500(105) (adopted on 28 April 2022) AMENDMENTS TO THE INTER NATIONAL MARITIME SOLID BULK CARGOES (IMSBC) CODE - 8 - .2 not more than 80% ammonium ni trate mixed with calcium carbonate and/or dolomite and/or mineral calcium sulphate and n ot more than 0.4% total combustible organic material calculated as carbon; or .3 mixture s of ammonium nitrate and ammonium sulphate with not more than 45% ammonium nitrate and not more than 0.

4% total combustible organic material calculated as carbon.

Compound NPK/NK/NP fertilizers .1 mixtures of nitrogen with phosphate and/or potash cont aining not more than 70% ammonium nitrate and not more than 0.4% total combustible --- U RL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCRe solutions/MSC.500(105).pdf .1 mixtures of nitrogen with phosphate and/or potash containi ng not more than 70% ammonium nitrate and not more than 0.4% total combustible organic m aterial calculated as carbon or not more than 45% ammonium nitrate and unrestricted comb ustible material; and .2 either less than 20% of ammonium nitrate content or less than 2 % of chloride (as indicated in the grey area of the figure for NPK/NP/NK fertilizers bel ow).

The shipper shall declare the ammonium nitrate content and the chloride content in accor dance with 4.2 of this Code.

Notwithstanding the above, fertilizers within these composition limits are not subject t o the provisions of this schedule, if they are assigned class 9 due to the hazard of sel f -sustaining decomposition based on the results of the trough test (referred to in the UN Manual of Tests and Criteria, part III, section 39).

Description Crystals, granules or prills.

Non-cohesive when dry.

Wholly or partly soluble in water.

Common --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResol utions/MSCResolutions/MSC.500(105).pdf and Criteria, part III, section 39).

Description Crystals, granules or prills.

Non-cohesive when dry.

Wholly or partly soluble in water.

Common products, listed (non-exhaustive) under this schedule are: .1 calcium ammonium ni trate; .2 ammonium sulphate nitrate; .3 ammonium nitrate with other sulphates (e.g.

calcium or magnesium sulphate); and .4 compound NPK/NP/NK fertilizer.

% Ammonium nitrate % Chloride 70 % 20 % 2 % .....

....

RESOLUTION MSC.500(105) (adopted on 28 April 2022) AMENDMENTS TO THE INTERNATIONAL MARIT IME SOLID BULK CARGOES (IMSBC) CODE - 9 - Characteristics Physical properties Size Angle of repose Bulk density (kg/m3) Stowage factor (m3/t) 1 to 5 mm 27° to 42° 1,000 to 1,200 0.83 to 1.00 Hazard classification Class Subsidiary hazard(s) MHB Group Not applicable N ot applicable Not applicable C Hazard When this cargo is heated strongly, it will decomp ose and give off toxic gases with the risk of --- URL: https://wwwcdn.imo.org/localresou rces/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.500(105).pdf Hazard Whe n this cargo is heated strongly, it will decompose and give off toxic gases with the ris k of toxic fumes in the cargo hold, adjacent spaces and on deck.

If decomposition is initiated in a localized area, it is highly unlikely to spread throu
ghout the mass of the fertilizer.

Fertilizer dust might be irritating to skin and mucous membranes.

It is a hygroscopic cargo and will cake if wet.

Stowage and segregation "Separated from" sources of heat.

Hold cleanliness Clean and dry as relevant to the hazards of the cargo.

Weather precautions This cargo shall be kept as dry as practicable.

This cargo shall not be handled during precipitation.

During handling of this cargo, all non -working hatches of the cargo spaces into which t
his cargo is loaded, or to be loaded, shall be closed.

Loading Trim in accordance with the relevant provisions required under sections 4 and 5
of this Code.

Precautions --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMO
Resolutions/MSCResolutions/MSC.500(105).pdf Loading Trim in accordance with the relevant
provisions required under sections 4 and 5 of this Code.

Precautions "NO SMOKING" signs shall be displayed on deck whenever this cargo is on boar
d.

Smoking shall not be allowed on deck and in the cargo spaces.

No welding, burning, cutting or other operations involving the use of fire, open flame,
spark or arc-producing equipment shall be carried out on equipment or structures in dire
ct contact with the fertilizer.

In order to avoid heating the cargo, all electrical equipment or other equipmen t capabl
e of developing heat, other than that of approved safe type, in the cargo spaces to be u
sed for this cargo shall be electrically disconnected from the power source, by appropri
ate means other than a fuse, at a point external to the space.

This situation shall be maintained while the cargo is on board.

Appropriate precautions shall be taken to protect machinery and accommodation spaces fro
m the dust of the cargo.

--- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/
MSCResolutions/MSC.500(105).pdf is on board.

Appropriate precautions shall be taken to protect machinery and accommodation spaces fro
m the dust of the cargo.

Bilge wells of the cargo spaces shall be protected from ingress of the cargo.

Due consideration shall be given to protect equipment from the dust of the cargo.

Persons who may be exposed to the dust of the cargo shall wear goggles or other equivale
nt dust eye -protection and dust filter masks.

Those persons shall wear protective clothing, as necessary.

RESOLUTION MSC.500(105) (adopted on 28 April 2022) AMENDMENTS TO THE INTERNATIONAL MARIT
IME SOLID BULK CARGOES (IMSBC) CODE - 10 - Ventilation The cargo spaces carrying this ca
rgo shall not be ventilated during voyage, except in an emergency.

Carriage No special requirements.

Discharge This cargo is hygroscopic and may cake in overhangs, impairing safety during d
ischarge.

If this cargo has hardened, it shall be trimmed to avoid the formation of overhangs, as
necessary.

Clean-up --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMORes
olutions/MSCResolutions/MSC.500(105).pdf If this cargo has hardened, it shall be trimmed
to avoid the formation of overhangs, as necessary.

Clean-up After discharge of this cargo, the bilge wells and the scuppers of the cargo sp
aces shall be checked and any blockage in the bilge wells and the scuppers shall be remo
ved." "AMMONIUM NITRATE BASED FERTILIZER MHB Ammonium nitrate based fertilizers transpor
ted under conditions mentioned in this schedule are uniform mixtures of nitrogen with or
without potash and/or phosphate within th e following composition limits: .1 not more th
an 70% ammonium nitrate and not more than 0.4% total combustible organic material calcul
ated as carbon or not more than 45% ammonium nitrate and unrestricted combustible materi

al; and .2 both the ammonium nitrate content is equal to or greater than 20% and the chl oride content is equal to or greater than 2% (as indicated in the grey area of the figur e below).

--- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/ MSCResolutions/MSC.500(105).pdf chloride content is equal to or greater than 2% (as indi cated in the grey area of the figure below).

The shipper shall declare the ammonium nitrate content and the chloride content in accor dance with 4.2 of this Code.

Notwithstanding the above, fertilizers within these composition limits are not subject t o the provisions of this schedule, if they are assigned class 9 due to the hazard of sel f -sustaining decomposition based on the results o f the trough test (referred to in the UN Manual of Tests and Criteria, part III, section 39).

% Ammonium nitrate % Chloride 70 % 20 % 2 % RESOLUTION MSC.500(105) (adopted on 28 April 2022) AMENDMENTS TO THE INTERNATIONAL MARITIME SOLID BULK CARGOES (IMSBC) CODE - 11 - De scription Crystals, granules or prills.

Non -cohesive when dry.

Wholly or partly soluble in water.

Common products listed under this schedule are compound NPK/NK fertilizers.

Characteristics Physical properties --- URL: https://wwwcdn.imo.org/localresources/en/Kn owledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.500(105).pdf Common products list ed under this schedule are compound NPK/NK fertilizers.

Characteristics Physical properties Size Angle of repose Bulk density (kg/m3) Stowage fa ctor (m3/t) 1 to 5 mm 27° to 42° 1,000 to 1,200 0.83 to 1.00 Hazard classification Class Subsidiary hazard(s) MHB Group Not applicable Not applicable OH B Hazard Although this c argo passes the trough test (referred to in the UN Manual of Tests and Criteria, part II I, section 39), and hence does not fall in class 9, when carried in bulk in large quanti ties, it may still be subject to decomposition if strongly heated from external sources.

Once initiated, decomposition might gradually spread through the remainder of the cargo, producing large volumes of toxic gases.

This cargo is not subject to an explosion hazard.

Fertilizer dust might be irritating to skin and mucous membranes.

It is hygroscopic cargo and will cake if wet.

Stowage and segregation --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentr e/IndexofIMOResolutions/MSCResolutions/MSC.500(105).pdf Fertilizer dust might be irritat ing to skin and mucous membranes.

It is hygroscopic cargo and will cake if wet.

Stowage and segregation ■Separated from■ sources of heat (see also Loading).

Not to be stowed immediately adjacent to any tank, double bottom or pipe containing heat ed fuel oil, unless there are permanent means and procedures to monitor and control the temperature so that it does not exceed 50°C.

Fertilizers of this type shall be stowed out of direct contact with a metal engine-room boundary.

This may be done, for example, by using flame-retardant bags containing inert materials or by any equivalent arrangement approved by the competent authority of the country of o rigin.

This requirement does not apply if the bulkhead is class A-60 or to short international voyages.

The hatches of the cargo spaces, including those of 'tween decks, shall be kept free at all times.

In case of an emergency, whenever this material is on board, opening the hatches must -- - URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MS CResolutions/MSC.500(105).pdf times.

In case of an emergency, whenever this material is on board, opening the hatches must be enabled (see 9.3.1.13 of this Code).

Hold cleanliness Clean and dry as relevant to the hazards of the cargo.

Weather precautions This cargo shall be kept as dry as practicable.

This cargo shall not be handled during precipitation.

During handling of this cargo, all non -working hatches of the cargo spaces into which t
his cargo is loaded, or is to be loaded, shall be closed.

Loading Trim in accordance with the relevant provisions required under sections 4 and 5
of this Code.

Prior to loading, the following provisions shall be complied with In order to avoid heat
ing up of the cargo, all electrical equipment or other equipment capable of developing h
eat, other than that of approved safe type, in the cargo spaces to be used for this carg
o, shall be electrically disconnected from the power source, by appropriate means --- UR
L: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCRes
olutions/MSC.500(105).pdf this cargo, shall be electrically disconnected from the power
source, by appropriate means other than a fuse, at a point external to the space.

This situation shall be maintained while the cargo is on board.

RESOLUTION MSC.500(105) (adopted on 28 April 2022) AMENDMENTS TO THE INTERNATIONAL MARIT
IME SOLID BULK CARGOES (IMSBC) CODE - 12 - Due consideration shall be given to the neces
sity to open hatches in case of decomposition to provide maximum ventilation, release pr
essure and heat, and slow down the reaction.

During loading, the following provisions shall be complied with Organic contamination ag
gravates decomposition behaviour in the presence of sources of heat, and therefore bunke
ring of fuel oil shall not be allowed during loading.

Pumping of fuel oil in spaces adjacent to the cargo spaces for this cargo, other than th
e engine-room, shall not be allowed.

Precautions ■NO SMOKING■ signs shall be displayed on deck whenever this cargo is on boar
d.

Smoking --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOReso
lutions/MSCResolutions/MSC.500(105).pdf engine-room, shall not be allowed.

Precautions ■NO SMOKING■ signs shall be displayed on deck whenever this cargo is on boar
d.

Smoking shall not be allowed on deck and in the cargo spaces.

No welding, burning, cutting or other operations involving the use of fire, open flame,
spark or arc-producing equipment shall be carried out on equipment or structures in dire
ct contact with the fertilizer.

The master and officers are to note that the ship's fixed gas fire -fighting installatio
n will be ineffective on decompositions involving this cargo and must not be used.

If decomposition is identified, water must be applied without delay.

Injection to the seat of decomposition is the first control measure because it uses less
water and can be more effective in early decomposition stages.

Total flooding is the final control measure but can introduce stability and stress issue
s.

The consequential risk to the stability of the ship through fluidization of the --- URL:
https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolu
tions/MSC.500(105).pdf and stress issues.

The consequential risk to the stability of the ship through fluidization of the cargo mu
st be taken into account in both cases.

Application of water to the surface of the cargo is much less effective and can give a f
alse sense of safety.

Appropriate precautions shall be taken to protect machinery and accommodation spaces fro
m the dust of the cargo.

Bilge wells of the cargo spaces shall be protected from ingress of the cargo.

Due consideration shall be given to protect equipment from the dust of the cargo.

Persons who may be exposed to the dust of the cargo shall wear goggles or other equivale
nt dust eye-protection and dust filter masks.

Those persons shall wear protective clothing, as necessary.

Ventilation The cargo spaces carrying this cargo shall not be ventilated during the voya
ge, except in an emergency.

Carriage There shall be a daily monitoring, recording and assessment of the trends of th

e cargo --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOReso
lutions/MSCResolutions/MSC.500(105).pdf emergency.

Carriage There shall be a daily monitoring, recording and assessment of the trends of th
e cargo temperature and oxygen concentration in the cargo space(s) throughout the voyage
.

Increase of temperature and decrease of oxygen concentration give an early indication of
a decomposition.

In addition, should decomposition occur, the residue left after decomposition may have o
nly half the mass of the original cargo.

Due consideration shall be given to the effect of the loss of mass on the stability of t
he ship.

Discharge Organic contamination aggravates decomposition behaviour in the presence of so
urces of heat, and therefore bunkering of fuel oil shall not be allowed during discharge
.

Pumping of fuel oil in spaces adjacent to the cargo spaces for this cargo, other than th
e engine-room, shall not be allowed during discharge.

RESOLUTION MSC.500(105) (adopted on 28 April 2022) AMENDMENTS TO THE INTERNATIONAL MARIT
IME SOLID BULK CARGOES (IMSBC) CODE - 13 - --- URL: https://wwwcdn.imo.org/localresource
s/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.500(105).pdf RESOLUTION MS
C.500(105) (adopted on 28 April 2022) AMENDMENTS TO THE INTERNATIONAL MARITIME SOLID BUL
K CARGOES (IMSBC) CODE - 13 - This cargo is hygroscopic and may cake in overhangs, impai
ring safety during discharge.

If this cargo has hardened, it shall be trimmed to avoid the formation of overhangs, as
necessary.

Clean-up After discharge of this cargo, the bilge wells and the scuppers of the cargo sp
aces shall be checked, and any blockage in the bilge wells and the scuppers shall be rem
oved.

Emergency procedures Special emergency equipment to be carried Protective clothing (boot
s, gloves, coveralls and headgear) Self-contained breathing apparatus Emergency procedur
es Wear protective clothing and self-contained breathing apparatus Emergency action in t
he event of fire or decomposition Decomposition in a cargo space containing this materia
l : The master and officers are to --- URL: https://wwwcdn.imo.org/localresources/en/Kno
wledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.500(105).pdf Emergency action in t
he event of fire or decomposition Decomposition in a cargo space containing this materia
l : The master and officers are to note that the ship's fixed gas fire -fighting install
ation will b e ineffective on decompositions involving this cargo and must not be used.

If decomposition is identified, water must be applied without delay.

Injection to the seat of decomposition is the first control measure (e.g.

using Victor lance) because it uses le ss water and can be more effective in early decom
position stages.

Total flooding is the final control measure but can introduce stability and stress issue
s.

The consequential risk to the stability of the ship through fluidization of the cargo mu
st be taken into account in both cases.

Application of water to the surface of the cargo is much less effective and can give a f
alse sense of safety.

Fire in an adjacent cargo space : Heat transferred from fire in an adjacent space can ca
use --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResoluti
ons/MSCResolutions/MSC.500(105).pdf and can give a false sense of safety.

Fire in an adjacent cargo space : Heat transferred from fire in an adjacent space can ca
use the material to decompose with consequent evolution of toxic fumes.

Open hatches to provide maximum ventilation.

Dividing bulkheads should be cooled.

Medical first aid Refer to the Medical First Aid Guide (MFAG), as amended " "CLAM SHELL
This schedule shall only apply to whole clam shells.

Description This cargo is a by -product generated in the process of clam farming.

Dark grey to beige, granular in form, not soluble, solid and natural material.

Characteristics Physical properties Size Angle of repose Bulk density (kg/m3) Stowage fa
ctor (m3/t) 5 mm to 30 mm 34° 1,058 0.945 Hazard classification Class Subsidiary hazard(
s) MHB Group Not applicable Not applicable Not applicable C RESOLUTION MSC.500(105) (ado
pted on 28 April 2022) AMENDMENTS TO THE INTERNATIONAL MARITIME SOLID BULK CARGOES (IMSB
C) CODE - 14 - --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/Indexof
IMOResolutions/MSCResolutions/MSC.500(105).pdf RESOLUTION MSC.500(105) (adopted on 28 Ap
ril 2022) AMENDMENTS TO THE INTERNATIONAL MARITIME SOLID BULK CARGOES (IMSBC) CODE - 14
- Hazard No special hazards.

This cargo is non-combustible or has a low fire risk.

Stowage and segregation No special requirements.

Hold cleanliness No special requirements.

Weather precautions No special requirements.

Loading Trim in accordance with the relevant provisions required under sections 4 and 5
of this Code.

Precautions No special requirements.

Ventilation No special requirements.

Carriage No special requirements.

Discharge No special requirements.

Clean-up After discharge of this cargo, the cargo spaces and the bilge wells shall be sw
ept clean and then thoroughly washed out." "LEACH RESIDUE CONTAINING LEAD Description In
termediate by-product formed as a result of the hydrometallurgical production of zinc an
d/or --- URL: https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolut
ions/MSCResolutions/MSC.500(105).pdf "LEACH RESIDUE CONTAINING LEAD Description Intermed
iate by-product formed as a result of the hydrometallurgical production of zinc and/or z
inc compounds.

Insoluble grey to brown granular substance obtained during dissolution of zinc ores or c
oncentrate in sulphuric acid to produce zinc sulphate solutions after physical separatio
n such as flotation and filtration.

Characteristics Physical properties Size Angle of repose Bulk density (kg/m3) Stowage fa
ctor (m3/t) Fines up to 100 µm 45° to 52° 800 to 1,600 0.63 to 1.25 Hazard classificatio
n Class Subsidiary hazards MHB Group Not applicable TX and CR A and B Hazard This cargo
may liquefy if shipped at a moisture content in excess of its transportable moisture lim
it (TML).

See sections 7 and 8 of this Code.

Harmful if swallowed or inhaled.

RESOLUTION MSC.500(105) (adopted on 28 April 2022) AMENDMENTS TO THE INTERNATIONAL MARIT
IME SOLID BULK CARGOES (IMSBC) CODE --- URL: https://wwwcdn.imo.org/localresources/en/Kn
owledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.502(105).pdf RESOLUTION MSC.502(1
05) (adopted on 28 April 2022) AMENDMENTS TO THE CODE OF SAFETY FOR SPECIAL PURPOSE SHIP
S, 1983 (1983 SPS CODE) MSC 105/20/Add.1 Annex 9, page 1 I:\MSC\105\MSC 105-20-Add.1.doc
x ANNEX 9 RESOLUTION MSC.502(105) (adopted on 28 April 2022) AMENDMENTS TO THE CODE OF S
AFETY FOR SPECIAL PURPOSE SHIPS, 1983 (1983 SPS CODE) THE MARITIME SAFETY COMMITTEE, REC
ALLING Article 28(b) of the Convention on the International Maritime Organization concer
ning the functions of the Committee, RECALLING ALSO resolution A.534(13), by which the A
ssembly, at its thirteenth session, adopted the Code of Safety for Special Purpose Ships
, 1983 (1983 SPS Code), RECALLING FURTHER that the Assembly authorized the Committee to
amend the Code